温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Multiple Linear Regression II

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Review

A linear regression model with dummy variables

$$Income = 1 + 0.1 * Gender + 0.4 * HZ + 0.3 * NB + 0.2 * WZ$$

- Income: continuous variable, million RMB
- Gender: 1 if female, 0 if male
- City: categorical variable with four categories
  - Reference group: Shaoxing
  - HZ = 1 if Hangzhou, 0 otherwise
  - NB = 1 if Ningbo, 0 otherwise
  - WZ = 1 if Wenzhou, 0 otherwise

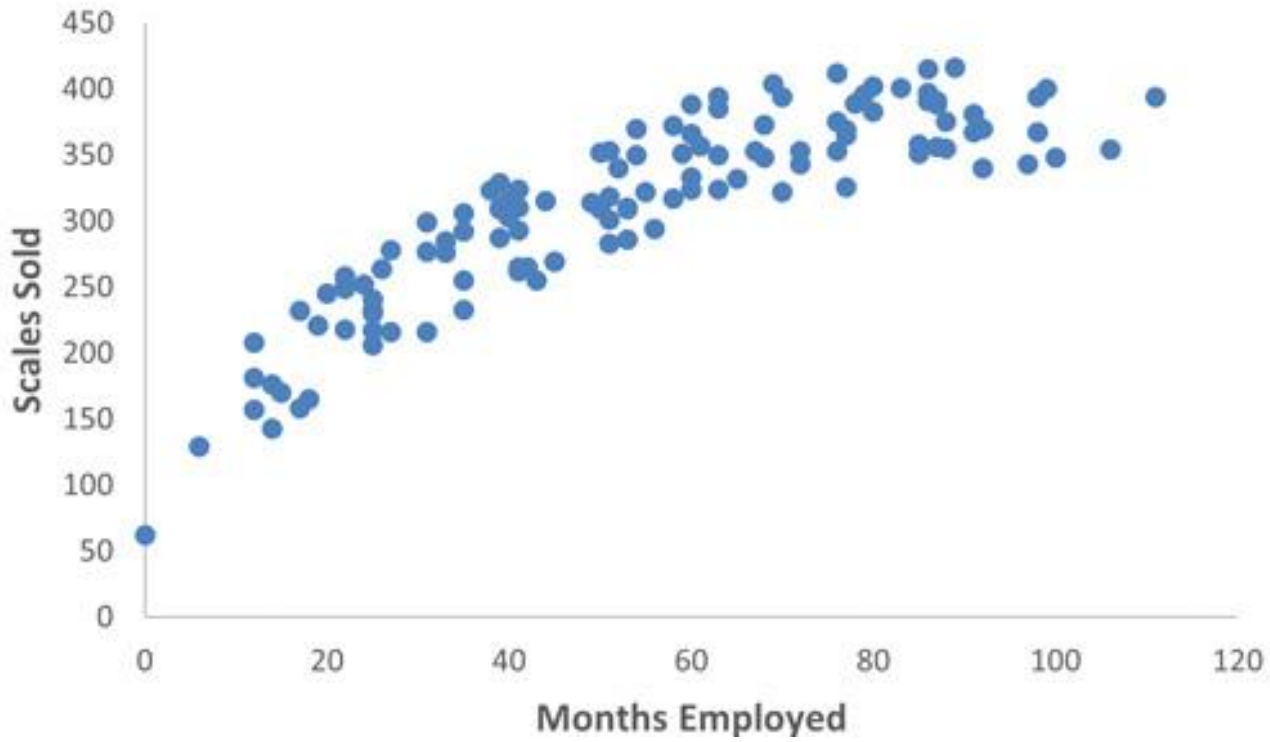
Think:

What is the average income difference between females in Hangzhou and males in Wenzhou?

Question: How would you interpret each regression coefficient?

Nonlinear Relationship

Context: A manager is examining how months of employment (x) relate to units sold (y) for 123 salespeople.



Question:

Based on the scatter plot, does the relationship between x and y appear **linear** or **curvilinear (nonlinear)**?

Nonlinear Relationship

The estimated simple linear regression equation:

$$Sales = 183.291 + 2.358 * Months$$

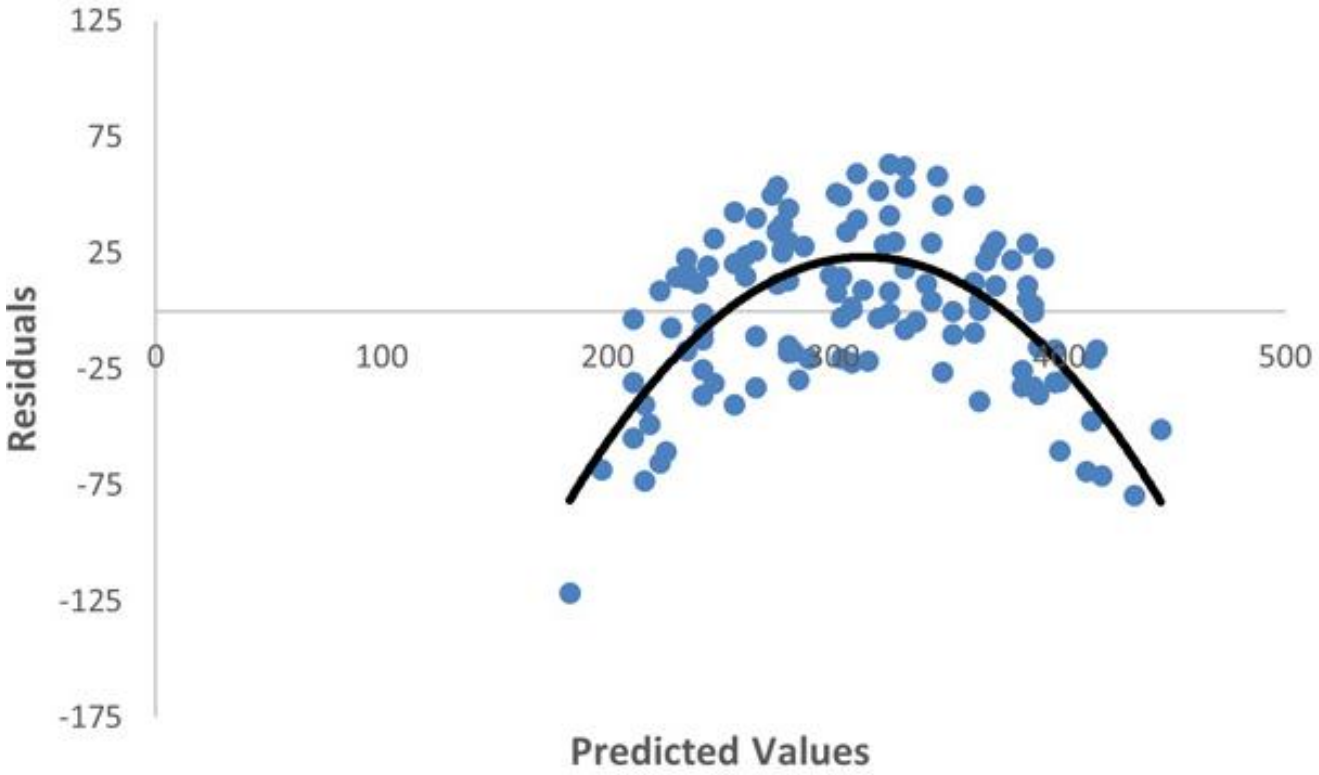
Regression Statistics	
R^2	0.756
Adjusted R^2	0.754
Observations	123

Questions:

- Is this a good estimation?
- Is it good enough?

	Coefficients	*Standard Error*	*t-stat*	*p-value*	*Lower 99%*	*Upper 99%*
Intercept	183.291	7.296	25.121	7.814E-50	164.196	202.387
Months Employed	2.358	0.122	19.383	6.530E-39	2.040	2.676

Nonlinear Relationship



Questions:

- What is on the x-axis?
- What is on the y-axis?
- Why does a nonlinear pattern between predicted values $\hat{\boldsymbol{y}}$ and residuals $\boldsymbol{\varepsilon}$ indicate a nonlinear relationship between $\boldsymbol{y}$ and $\boldsymbol{x}$?

Nonlinear Relationship

- To model the curvilinear relationship, we add a new variable $\boldsymbol{MonthSq} = (\boldsymbol{Months})^2$ and estimate a quadratic regression:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$$

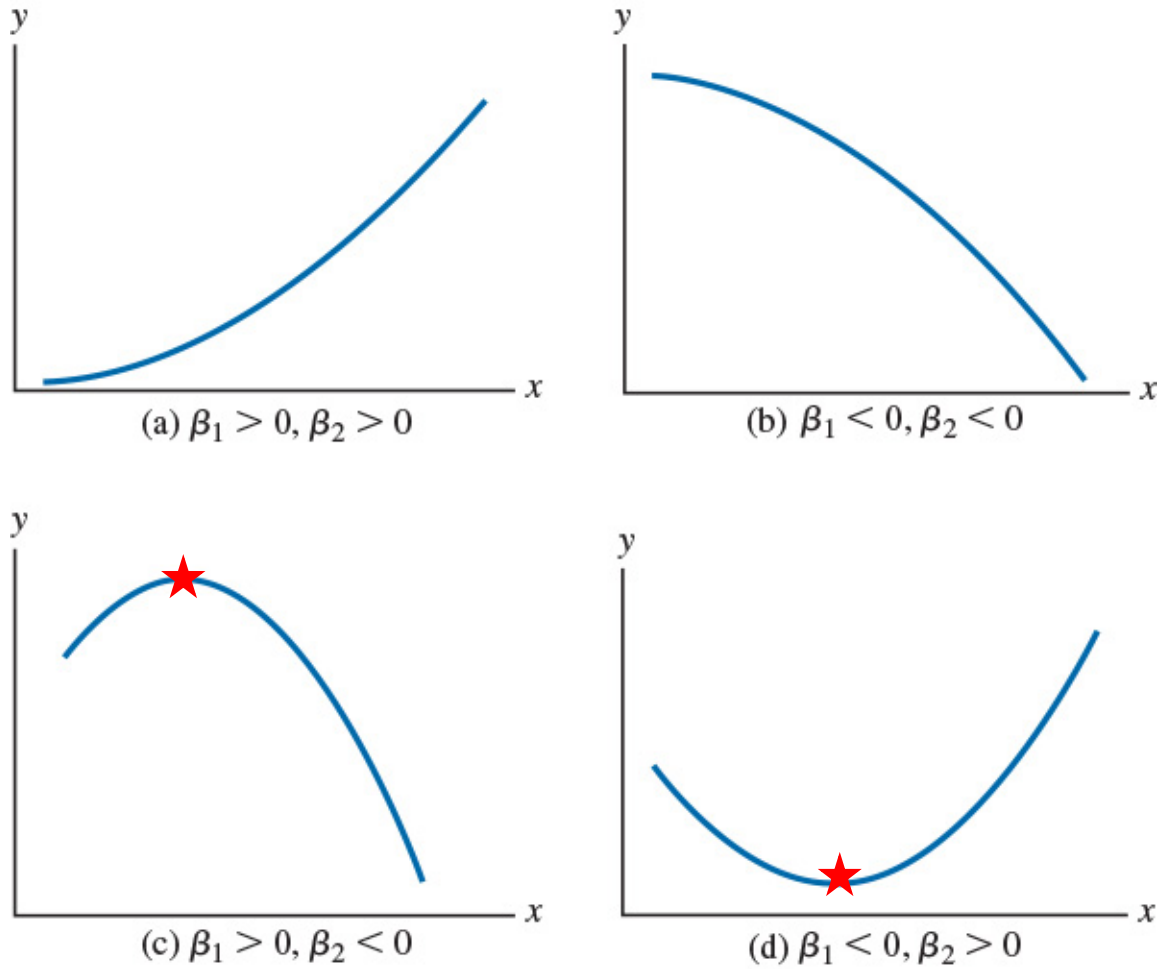
Regression Statistics	
R^2	0.865
Adjusted R^2	0.863
Observations	123

Question:
How do you interpret this quadratic relationship?

	Coefficients	*Standard Error*	*t-stat*	*p-value*
Intercept	101.985	9.882	10.320	2.932E-18
Months Employed	6.136	0.398	15.582	1.309E-30
MonthsSq	-0.034	0.003	-9.860	3.701E-17

Nonlinear Relationship

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$



- Experimental range of x
- Monotonically increasing / decreasing
- U-shape, inverse U-shape
- Maximum, minimum

Interaction Between IVs

- An interaction occurs when the effect of **one IV** on y depends on the value of **another IV**.
  - In other words, the effect of x_1 is not constant across levels of x_2, and vice versa.
  - Example: Train × Experience → training works better when experience is higher.

- To model this, include an interaction term:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2$$

$\hat{\beta}_3 > 0 \rightarrow$ Complementary

$\hat{\beta}_3 < 0 \rightarrow$ Substitute

- Interpretation notes:
  - $\hat{\beta}_3$: how the effect of one variable changes with the other variable.
  - $\hat{\beta}_1$ and $\hat{\beta}_2$: effects only when the other variable = 0.
  - Variables must be interpreted together, not separately.

How to interpret $\hat{\beta}_3$?

Interaction Between IVs

1) Main effects cannot be interpreted alone
   - $\hat{\beta}_1$: effect of x_1 only when $x_2 = 0$
   - $\hat{\beta}_2$: effect of x_2 only when $x_1 = 0$

Question:

What is the impact of a 1-unit increase in x_1?

2) Interactions introduce nonlinearity
   - Adding $x_1 x_2$ allow the slope of one variable to change with the other, even in a linear model.
3) Interactions increase multicollinearity
   - $x_1 x_2$ is mathematically related to x_1 and x_2, VIFs often increase.
   - This does not invalidate the model but can make coefficients less precise.
4) Always include main effects
   - If you include $x_1 x_2$, you should also include x_1 and x_2.
   - Dropping main effects while keeping an interaction leads to misleading interpretations.

Interaction Between IVs

5) Coefficient signs depend on coding
   - For dummy variables, the meaning of $\hat{\beta}_3$ depends on which category is coded as 1 (see next slides).
   - Re-coding changes the interpretation but not fitted values.

6) Plotting interactions is essential
   - Interpretations are easier to see with visuals (e.g., simple slopes, predicted values at different levels of the moderator).

7) Statistical significance matters
   - A positive or negative $\hat{\beta}_3$ still needs to be statistically significant.

Interaction Between IVs

Question:

Guess the sign of $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$

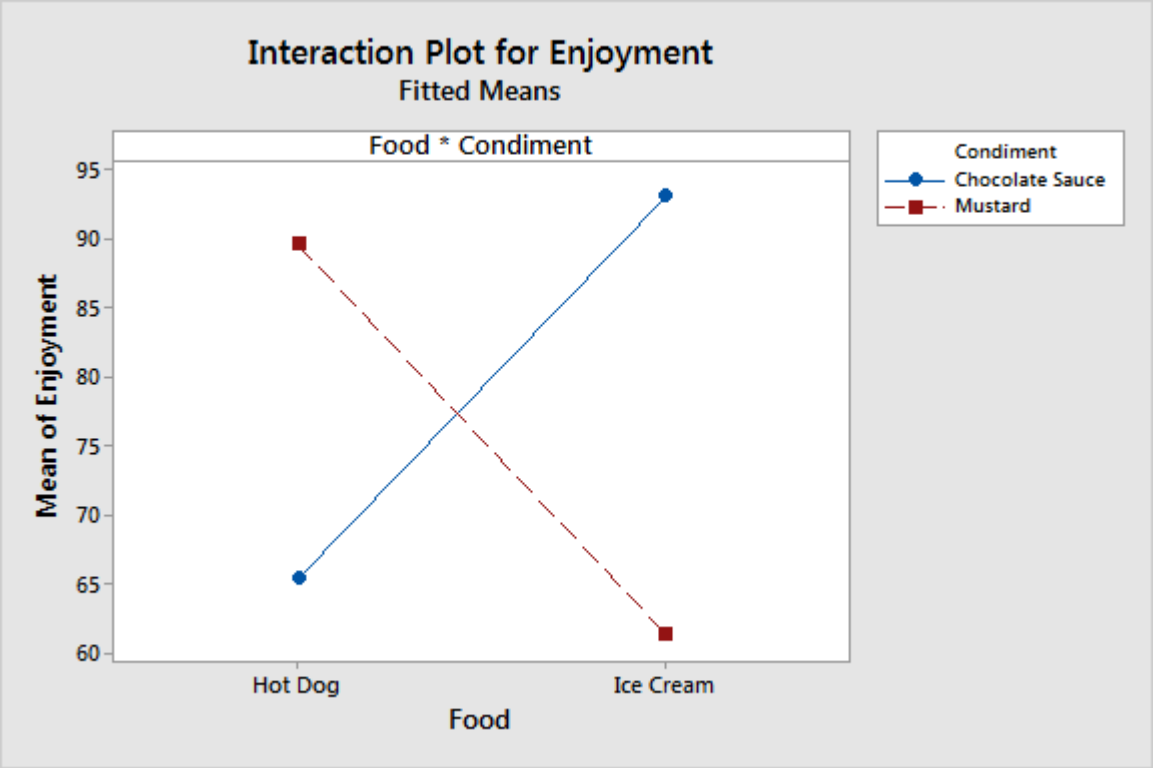
$$Enjoyment = \hat{\beta}_0 + \hat{\beta}_1 * Food + \hat{\beta}_2 * Condiment + \hat{\beta}_3 * Food * Condiment$$

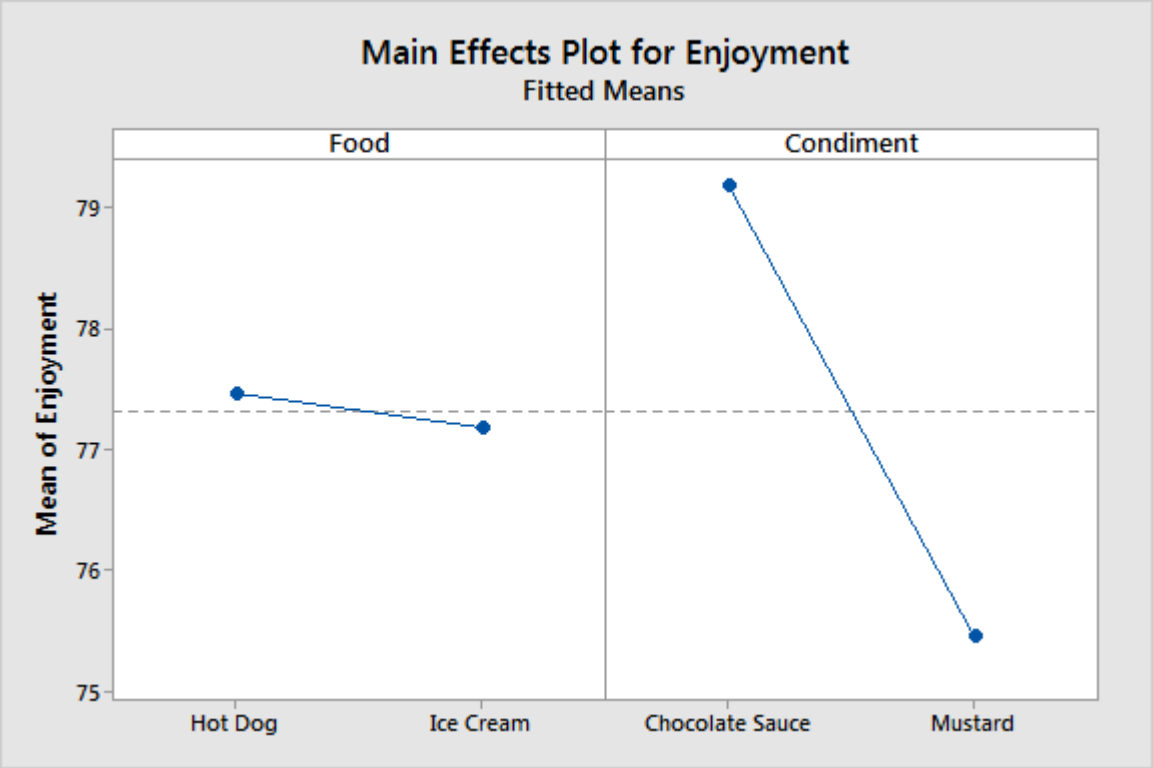
Term	***Coef***	***SE Coef***	***t-stat***	***p-value***	***VIF***
Constant	77.320	0.560	138.13	0.000	
Food					
Hot Dog	0.141	0.560	0.25	0.801	1.00
Condiment					
Chocolate Sauce	1.863	0.560	3.33	0.001	1.00
Food∗Condiment					
Hot Dog∗Chocolate Sauce	-14.007	0.560	-25.02	0.000	1.00

https://statisticsbyjim.com/regression/interaction-effects/

Interaction Between IVs

$$Enjoyment = \widehat{\beta}_0 + \hat{\beta}_1 * Food + \hat{\beta}_2 * Condiment + \hat{\beta}_3 * Food * Condiment$$





https://statisticsbyjim.com/regression/interaction-effects/

Variable Selection

- Why
  - Multiple regression often has many potential predictors, but not all add value.
  - Some variables add noise, increase multicollinearity, or reduce clarity.
- Goal
  - Build a model that is **interpretable**, **accurate**, **parsimonious** (no unnecessary variables)
- Risks of including too many variables
  - Overfitting: model fits noise (training data), not true patterns (sample data)
  - Multicollinearity: unstable and imprecise coefficients
  - Reduced interpretability: harder to explain relationships
  - Higher data requirements: more parameters = more uncertainty

Overfitting

- Overfitting occurs when the model is **too complex** and captures noise rather than true patterns.
- Such a model performs very well on the sample data but poorly on new data.
- How to avoid overfitting
  - Include only meaningful IVs
  - Use complex models **only when justified**
  - Treat variable selection procedures as guidance, not rules
  - If enough data, use **cross-validation** to test performance on unseen data
    - Example: holdout method (split into training and validation sets)

Variable Selection

- Criteria for including a variable
  - Theoretical justification (**<u>most important</u>**)
  - Statistical significance: p-values, t-stat, confidence intervals
  - Model fit: adjusted R^2 (penalizes unnecessary IVs)
  - Multicollinearity: check VIF
  - Practical significance: effect size and direction, business relevance
- Common variable selection methods
  - **Forward selection**: start with no variables, add the most significant variable one at a time, stop when no improvement
  - **Backward elimination**: start with all variables, remove the least significant variable step by step
  - **Stepwise selection**: combination of forward + backward, adds and removes variables iteratively

Variable Selection

- Practical guidelines
  - Start with theory-driven variables
  - Add statistical tools (p-value, adjusted R^2, VIF) to refine
  - Avoid automatic selection without thinking
  - Prefer a simpler model when performance is similar
- Key takeaways
  - Variable selection balances fit, parsimony, and interpretability
  - Use theory first, statistics second
  - Evaluate models holistically, not by one metric
  - A good model answers the business question clearly and reliably

What does a significant regression coefficient mean?

Significant at 0.1/0.05/0.01 level

Statistical Significance, p-value

Table 4. Change in Number of Requested Third-Party Domains and Cookies

	Requests		Cookies	
	(1) EU firm	(2) Non-EU firm	(3) EU firm	(4) Non-EU firm
Post × *EU audience*	−0.0789***	−0.0464***	−0.1096***	−0.0722***
	(0.0026)	(0.0028)	(0.0036)	(0.0042)
Post × *non-EU audience*	−0.0504***	−0.0220***	−0.0674***	−0.0310***
	(0.0038)	(0.0011)	(0.0051)	(0.0015)
Observations	961,587	2,691,711	961,587	2,691,711
$\overline{R^2}$	0.8871	0.9054	0.8444	0.8641
Mean dependent variable	2.3706	2.4647	1.3476	1.4296

Notes. Dependent variable in columns (1) and (2) is the $\log(y+1)$ number of third-party domains that the website-host requests and in columns (3) and (4) the $\log(y+1)$ number of third-party domains that the website-host requests and respond with a cookie. *Post* indicates the period after May 25, 2018. *EU Audience* indicates whether a website host has a top-level domain that is specific to a country in the EU and/or whether the website's language is in a EU language and/or whether the website appears on any EU country's Alexa ranking. *Non-EU Audience* indicates the opposite. *EU firm* indicates that the website domain is associated with a firm that lists a headquarter in the EU on Crunchbase and/or a website where the Whois records indicate an owner with an address in the EU. Separate linear time trend for EU audience/non-EU audience before/after and website host fixed effects in all specifications. Standard errors in parentheses, clustered on the website-host-level.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Regulatory Spillovers and Data Governance: Evidence from the GDPR

Significance Increases with Sample Size

- Standard errors shrink as sample size increases
  - Smaller SE → larger t-value → smaller p-value → more “significant” results
- Noise averages out
  - More data reduces random variation, making estimates more precise and stable
- Hypothesis tests become more sensitive
  - With big n, even slight deviations from the null hypothesis are detected
- Large samples detect tiny effects
  - Even very small or practically unimportant coefficients can become statistically significant
- Key point
  - Statistical significance becomes easier with large samples, but significance ≠ practical importance
  - Focus on **effect size**, **confidence intervals**, and **business relevance**, not just p-value

The Significant Problem of p-value

- p-hacking, data dredging / fishing / snooping
  - The manipulation of data analysis until it produces appear statistically significant results, compromising the truthfulness of the findings.
- Why p-values can mislead
  - A p-value measures how unlikely the sample data are **if the null hypothesis were true**, not the probability the hypothesis is true.
  - Large samples: even tiny and unimportant effects can become “statistically significant.”
  - Small samples: real effects may fail to reach significance.
- Better practices
  - Focus on effect sizes, confidence intervals, and practical significance
  - Encourage replication and transparent reporting
  - Use p-values as **one piece of evidence**, not the sole decision rule

Big little lies: a compendium and simulation of p-hacking strategies

Python Hands-On

Refer to the “demo_multiple_linear_regression.ipynb”

$$GPA = \beta_0 + \beta_1 * StudyHours + \beta_2 * StudyHours^2 + \beta_3 * SleepHours + \beta_4 * StudyHours * SleepHours + \varepsilon$$

Step 1: Create the population and sample

Step 2: Exploratory plots

Step 3: Model 1 – Linear effects only

Step 4: Model 2 – Add a quadratic term for StudyHours

Step 5: Model 3 – Quadratic + Interaction

Step 6: Compare the models

Summary

- **Nonlinear** patterns may require quadratic terms to better capture relationships.
- **Interaction** effects show how one variable's influence depends on another; interpret coefficients jointly.
- **Variable selection** aims for models that are interpretable, accurate, and parsimonious.
- Avoid **overfitting** by using meaningful variables, keeping models simple, and applying cross-validation when possible.
- **p-values** can mislead; avoid p-hacking and emphasize effect sizes, confidence intervals, and practical significance.
- A strong regression model should answer the business question *clearly, credibly, and reliably*.